

ENGR 1451/2451: Exploratory Data Science

More about Correlation

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Today's Goals

Use correlation carefully: know what changes it, what does not, and when it can mislead.

- Explain why correlation is unitless and unchanged by simple unit conversions.
- Predict how swapping variables, shifting, or rescaling affects r .
- Recognize outliers and nonlinear patterns where r is a poor summary.
- Distinguish individual-level correlations from ecological correlations.
- Explain why association alone does not establish causation.

Warm-Up: What Should Happen to r ?

A small data set has $r \approx 0.76$.

x	y
1	2
2	3
3	1
4	5
5	6

Question

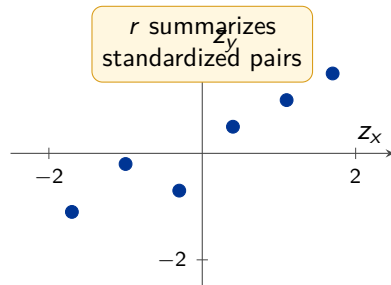
What happens to r if we...

1. switch the two columns?
2. add 3 to every y value?
3. double every x value?
4. multiply every y value by -1 ?

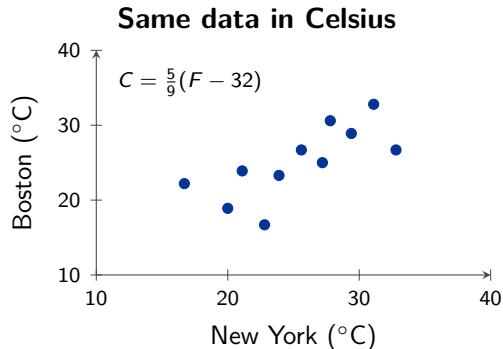
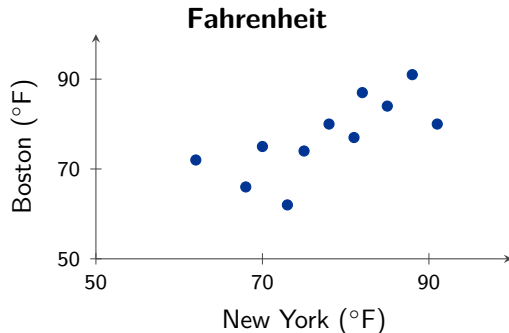
Correlation Is a Pure Number

Correlation is computed after converting both variables to standard units:

$$z_x = \frac{x - \bar{x}}{s_x}, \quad z_y = \frac{y - \bar{y}}{s_y}.$$

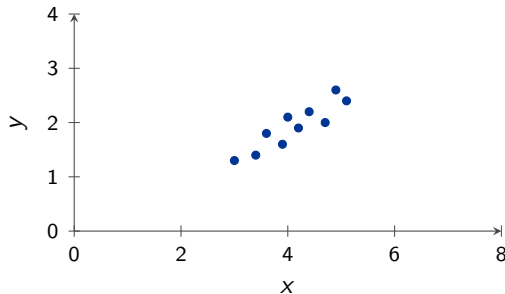


Unit Conversions Do Not Change r

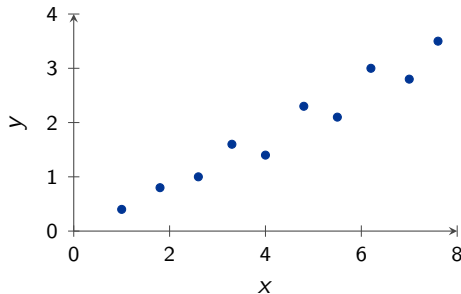


Same r Can Look Different

Narrow spread in both variables



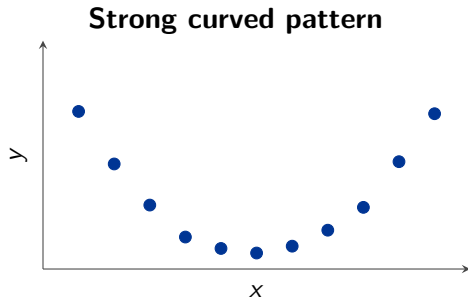
Wider spread in both variables



Idea

Correlation describes clustering **relative to the variables' spreads**, not in the original measurement units.

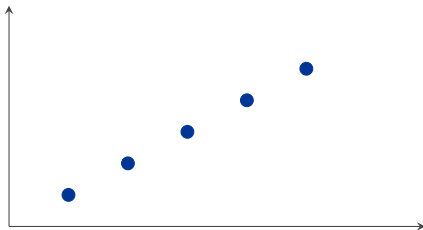
Correlation Measures Linear Association



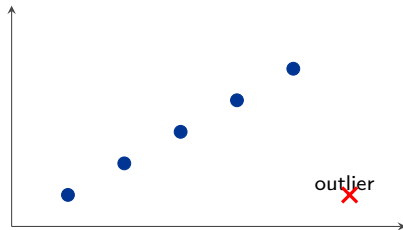
- r summarizes straight-line association.
- A curved pattern can have $r \approx 0$.
- Do not translate “small r ” into “no relationship” without seeing the plot.

Outliers Can Dominate r

Nearly perfect linear pattern



Same pattern plus one outlier



Better practice

Investigate unusual points. Remove a point only with a good reason, not just because it changes the answer.

Ecological Correlations

An **ecological correlation** is computed from group rates or group averages.

Individual level

Each point is a person, item, product, sample, etc.

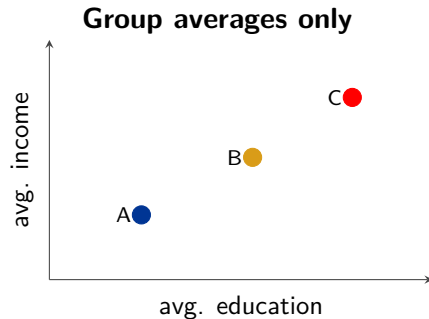
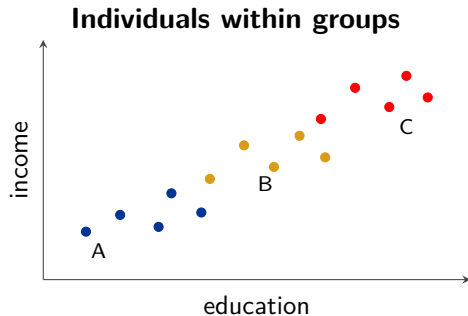
Group level

Each point is a state, country, classroom, year, school, company, etc.

- Group averages often look less scattered.
- The group-level correlation can be much larger.
- It may not estimate the individual-level relationship.

Same variables. Different unit of analysis.

Individuals Versus Group Averages



Watch out

Replacing individuals by averages removes much of the within-group scatter. Ecological correlations often overstate the strength of an association.

Quick Check: Ecological or Individual?

Question

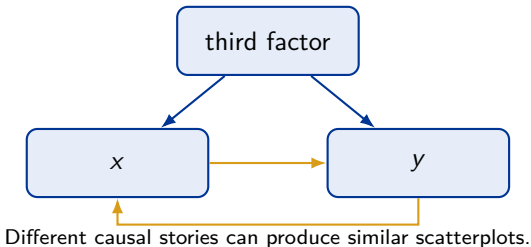
For each data set, is the correlation ecological or individual-level?

1. Each row is one student: study hours and exam score.
2. Each row is one class section: average study hours and average exam score.
3. Each row is one country: per-capita cigarette consumption and lung-cancer death rate.
4. Each row is one product: thickness and measured strength.

Association Is Not Causation

Correlation measures association. It does not identify the mechanism.

- x may cause y .
- y may cause x .
- A third factor may affect both.
- The pattern may be an artifact of measurement, sampling, or grouping.

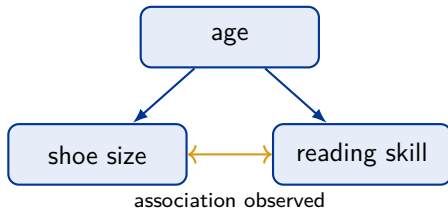


Confounding Example: Shoe Size and Reading

Among school children, shoe size and reading skill may be strongly associated.

Does reading make feet grow?

No. Age is a third factor: older children tend to read better and have larger feet.



Example: Education and Unemployment

During the Great Depression, better-educated job seekers tended to have shorter spells of unemployment.

Tempting conclusion

Education protected workers from unemployment.

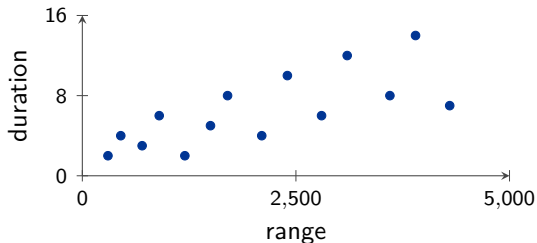
Caution

The data were observational. Age was a confounder: younger people tended to be better educated, and employers tended to prefer younger job seekers.

Example: Species Range and Duration

Observed association

For 99 gastropod species, geographical range and species duration had a positive association.



1. Wide range may help a species survive longer.
2. A long-lived species may have more time to spread.
3. Species with wider range may be easier to find in the fossil record.

Example: Diet and Cancer Across Countries

Across countries, higher per-capita fat consumption was associated with higher breast-cancer death rates.

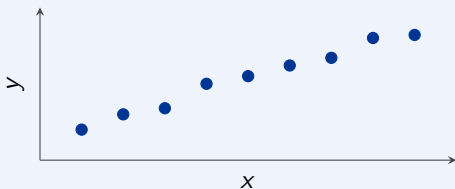
Why this is not enough

The points are countries, not people. Other country-level differences may move with diet.

- Richer countries may have more fat *and* more sugar.
- Lifestyle, screening, reporting, and other diet factors may differ.
- An ecological pattern can suggest a question, but does not settle causation.

Causal Explanations to Consider

Observed association



Before claiming causation, ask:

1. What was the study design?
2. Could a third variable explain it?
3. Could the direction be reversed?
4. Could measurement, selection, or aggregation create the pattern?

Key Takeaways

- Correlation is unitless because it is based on standardized values.
- Switching variables, adding constants, and multiplying by positive constants do not change r .
- Correlation measures linear association; nonlinear patterns and outliers can make it misleading.
- Ecological correlations use group rates or averages and can overstate individual-level association.
- Association is not causation; look for confounders, reverse direction, and study-design limits.